



Rajiv Gandhi University of Knowledge Technologies- AP.

(Catering to the Educational Needs of Gifted Rural Youth of Andhra Pradesh)

PROBABILITY AND SATATISTICS

Probability and theorems in Probability

- 1 **Basic Concepts in Probability:** Experiment, Sample Space (Discrete and Continuous), Event, Exhaustive Events, Favorable event, Impossible Event, Mutually exclusive Event, Equally likely Events.
- 2 **Definitions of Probability:** Mathematical Definition of Probability (Classical Probability), Statistical Definition of Probability (Probability as a Relative Frequency), Subjective definition of probability, Axiomatic Definition of Probability and solved problems on mathematical definition of probability.
- 3 **Theorems in Probability:** Joint Probability (Addition Theorem of Probability for two Events) and solved problems, Conditional probability, Independent events, Multiplication Theorem for Two events and solved problems.
- 4 **Bayes theorem and Its applications:** Total Probability, Bayes Theorem (Proof) and solved problems.

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(I) Basic Concepts in Probability:

Experiment: An Experiment is any physical action or process.

Ex:

- ❖ Tossing a coin
- ❖ Throwing a die
- ❖ Sitting in the balcony of the house and watching the movement of clouds in the sky is an experiment.

Experiments are classified into two types

- (1) Deterministic or Predictable Experiment.
- (2) Probabilistic or Unpredictable or Random Experiment.

(1) Deterministic experiment: If an experiment produces the same result when it is conducted several times under identical conditions, then the experiment is known as determinant experiment.

Note: All the experiments in physical and engineering sciences are deterministic.

Example:

- 1: If r is the radius of the sphere, then volume of sphere is $V = \frac{4}{3} \pi r^3$ which is deterministic (unique)
- 2: Throwing a stone upward when it is known that stone will definitely fall to the ground due to the force of gravitation.
- 3: Measuring the length of the table (or) black board

(2) Random Experiment: If an experiment produces different results even though it is conducted several times under identical conditions, then the experiment is known as random experiment. i.e.:

- a) All the outcomes of the experiment are known in advance
- b) The result is not knowing in advance
- c) The experiment can be repeated under identical conditions.

Note: All the experiments in social sciences are random.

Example:

- 1: A coin is tossed, it is known that either head or tail will appear but it is not known that exactly what the outcome will be, and the coin may be a tossed any number of times.
- 2: By looking at the sky, one is not sure whether the rain comes or not, in such cases we talk of chances or probability which can be taken as a Quantitative measure of certainty.

Sample Space & sample point: The set of all possible outcomes of a random experiment is called Sample Space, and is denoted by S . Each outcome of a random experiment is called sample point.

Example:

- a) In tossing of a coin there are two possible outcomes. Sample space $S = \{H, T\}$ and $n(s)=2$
- b) In throwing a die there are 6 possible outcomes. Sample space $S = \{1, 2, 3, 4, 5, 6\}$ and $n(s)=6$

c) In drawing two cards from pack of cards then possible outcomes $n(s)=52C_2$

Note:

- If n coins are tossed, then the number of elements in the sample space is $n(s)= 2^n$.
- If n dice are thrown (rolled) then the number of elements in sample space is $n(s)= 6^n$.
- The number of ways selecting “ r ” things from “ n ” thing is $n(s)= nCr$

Discrete Sample Space: if a sample space contains finite or countable number of elements then it is said to be discrete sample space.

Ex: In tossing a coin the sample space $S = \{H, T\}$ and $n(s)=2$ (finite)

Continuous Sample Space: if a sample space contains infinite or uncountable number of elements then it is called as continuous sample space.

Ex: Consider a random experiment of tossing a coin until head appears then sample space $S=\{H, TH, TTH, TTTH, TTTTH, \dots\}$ and $n(s)=$ infinite. Here TTH represents tail in first, tail in second and head in third tosses and so on.

Trial: Conducting a random experiment once is known as a trial

Event: A subset of a sample space is known as an event. Events are denoted by $E_1, E_2, \dots, E_n$

Example:

- a) Two coins are tossed in an experiment and getting HT (one head and one tail) is event
- b) Throwing a die two times is an experiment, getting their sum is 10 is event

Exhaustive Events: The events in a sample space are said to be exhaustive if their union is equal to the sample space. The events $E_1, E_2, E_3, \dots, E_n$ in S are said to be exhaustive if $\bigcup_{i=1}^n E_i = S$ i.e; The total number of possible outcomes in experiment is called exhaustive event and is denoted by $n(s)$

Favorable event: The number of cases favorable to an event in a trial is the no of outcomes which entire the happening of the event.

Ex: In throwing of two dice, the number of cases favorable for getting their sum 10 is 3 i.e.; (4,6), (5,5), (6,4). Hence $n(\text{favorable for getting their sum } 10)= 3$

Impossible Event: An event with no outcome in it is known as impossible event and is denoted by ϕ .

Mutually exclusive Event: If two or more of them cannot happen simultaneously in the same trial then the event is called mutually exclusive event.

Ex: In throwing a dice experiment, the events 1,2,3,-----6 are M.E. events

Note: if E_1 and E_2 are exclusive events then $E_1 \cap E_2 = \emptyset$

Equally likely Events: Outcomes of events are said to be equally likely if there is no reason for one to be preferred over other.

Ex: In throwing a die, chance of getting 1, 2, 3, 4, 5, 6 are equally likely

Example1: In a random experiment of tossing three coins,

(a) The sample space is given by $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ and $n(s) = 2^3 = 8$

(b) Let us define some events in the sample space as below:

(i) Let E_1 be the Three heads

(ii) Let E_2 be the Three tails

(iii) Let E_3 be the Exactly one head or Exactly Two tails

(iv) Let E_4 be the Exactly Two heads or Exactly one tail

(v) Let E_5 be the at least one head

(vi) Let E_6 be the at least two heads

(vii) Let E_7 be the at most one head

(viii) Let E_8 be the at most two heads.

(ix) Let E_9 be the at most three heads

(x) Let E_{10} be the Exactly Four heads

(c) **Favorable events:**

(i) $E_1 = \{HHH\}$ and $n(E_1) = 1$

(ii) $E_2 = \{TTT\}$ and $n(E_2) = 1$

(iii) $E_3 = \{HTT, THT, TTH\}$, and $n(E_3) = 3$

(iv) $E_4 = \{HHT, HTH, THH\}$ and $n(E_4) = 3$

(v) $E_5 = \{HHH, HHT, HTH, HTT, THH, THT, TTH\}$ and $n(E_5) = 7$

(vi) $E_6 = \{HHT, HTH, THH, HHH\}$ and $n(E_6) = 4$

(vii) $E_7 = \{HTT, THT, TTH, TTT\}$, and $n(E_7) = 4$

(viii) $E_8 = \{HTT, THT, TTH, HHT, HTH, THH, TTT\}$ and $n(E_8) = 7$

(ix) $E_9 = \{HHT, HTH, HTT, THH, THT, TTH, HHH, TTT\}$ and $n(E_9) = 8$

(x) $E_{10} = \{\}$ and $n(E_{10}) = 0$

(d) **Exhaustive Events:** $E_1 \cup E_2 \cup E_3 \cup E_4 = S$, hence E_1, E_2, E_3 and E_4 are exhaustive events in S

(e) Mutually exclusive event: $E_i \cap E_j = \emptyset$, where $i \neq j$. Hence, E_1, E_2, E_3, E_4 are mutually exclusive events in S .

(f) Impossible event: E_{10} is impossible event.

Example2: In a random experiment of throwing two dice find

(a) sample space (b) Exhaustive Events (c) Mutually exclusive event (d) Impossible event

(II) Definitions of Probability:

(A) Mathematical Definition of Probability (Classical or A Priori Probability): In a random experiment let there be “n” exhaustive, mutually exclusive and equally likely events and let “E” be an event of the experiment and “m” of them are favorable to the happening of an event E then the probability of an event E is denoted by P(E) and is defined as

$$P(E) = \frac{\text{number of favorable events of E}}{\text{number of outcomes in S}} = \frac{n(E)}{n(S)} = \frac{m}{n}$$

If $\bar{E}$ denotes the non-occurring of E then the probability of $\bar{E}$ is $P(\bar{E}) = \frac{n-m}{n} = 1 - \frac{m}{n}$

$$P(\bar{E}) = 1 - P(E) \Rightarrow P(\bar{E}) + P(E) = 1$$

Note: Since m is non-negative integer and n is positive integer

$$\text{since } m \geq 0 \text{ and } n > 0 \Rightarrow 0 \leq m \leq n \Rightarrow 0 \leq \frac{m}{n} \leq 1 \Rightarrow 0 \leq P(E) \leq 1$$

Similarly, $0 \leq P(\bar{E}) \leq 1$

Limitations:

- (i) It is used only when outcomes are equally liked.
- (ii) It cannot be used the number of outcomes infinite

(B) Statistical Definition of Probability (Empirical or Relative Frequency or Von Mises):

If an experiment is performed repeatedly under essentially homogeneous and identical conditions, then the limiting value of the ratio of the number of times the event occurs to the number of trials.

Symbolically, if in ‘n’ trials an event E happens ‘m’ times, then the probability of the happening of ‘E’ denoted by P(E), is given by $P(E) = \lim_{n \rightarrow \infty} \frac{m}{n}$, Where $\frac{m}{n}$ is the relative frequency of event E

Limitations of Empirical Probability

- (i) If an experiment is repeated a large number of times, the experimental conditions may not remain identical.
- (ii) The limit may not attain a unique value, however large may be.

Ex: $m = \sqrt{n}$ then $P(E) = \lim_{n \rightarrow \infty} \frac{m}{n} = \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n} = 0$ Probability of rare events is zero but it does not mean that the event does not occur.

(C) Subjective definition of probability: In this method, probabilities are assigned to events according to the knowledge, experience and belief about the happening of the events. The main limitation of this definition is, it varies from person to person.

(D) Axiomatic Definition of Probability: Let S be the sample space and E be an event associated with a random experiment. Then the probability of event E occurring denoted by P(E) is defined as a real number satisfying the following axioms

a) $0 \leq P(E) \leq 1$ (For any event E is associated a real number between 0 and 1)

b) $P(S)=1$ (The sum of the probability of all events in a sample space is one)

c) If A & B are mutually exclusive events, then $P(A \cup B) = P(A) + P(B)$

Note: $P(\cup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} P(A_k)$ where $A_1, A_2, A_3, \dots, A_k$ are mutually disjoint events of S

1. If three coins are tossed at a time, then the probability of getting (i) exactly one Head (ii) exactly two Heads (iii) at least two Heads (iv) at most two Heads (v) at least one Head (vi) at most one Head.

Sol:

If three coins are tossed at a time, then the possible outcomes are $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ and $n(s) = 2^3 = 8$

(i) Let E_1 be an event getting exactly one head

The number of favorable events are getting exactly one head $E_1 = \{THT, TTH, HTT\}$ and $n(E_1) = 3$

Hence the probability of getting exactly one Head $P(E_1) = \frac{n(E_1)}{n(s)} = \frac{3}{8}$

(ii) Let E_2 be an event getting exactly two heads then $E_2 = \{HHT, HTH, THH\}$ and $n(E_2) = 3$

Probability of getting two heads $P(E_2) = \frac{n(E_2)}{n(s)} = \frac{3}{8}$

(iii) Let E_3 be an event getting at least two heads then $E_3 = \{HHT, HTH, THH, HHH\}$ and $n(E_3) = 4$

Probability of getting at least two heads $P(E_3) = \frac{n(E_3)}{n(s)} = \frac{4}{8}$

(iv) Let E_4 be an event getting at most two heads then $E_4 = \{HHT, HTH, THH, THT, TTH, HTT, TTT\}$ and $n(E_4) = 7$

Probability of getting at most two heads $P(E_4) = \frac{n(E_4)}{n(s)} = \frac{7}{8}$

(v) Let E_5 be an event getting at least one head then $E_5 = \{HHT, HTH, THH, THT, TTH, HTT, HHH\}$ and $n(E_5) = 7$

Probability of getting at least one head $P(E_5) = \frac{n(E_5)}{n(s)} = \frac{7}{8}$

(vi) Let E_6 be an event getting at most one head then $E_6 = \{THT, TTH, HTT, TTT\}$ and $n(E_6) = 4$

Probability of getting at most one head $P(E_6) = \frac{n(E_6)}{n(s)} = \frac{4}{8}$

2. A uniform die is thrown at random. Find the probability that the number on it is (i) even (ii) odd (iii) even or multiple of 3 (iv) even and multiple of 3 (v) greater than 4. (vi) a composite number

Sol:

Let S be the sample space of throwing a die and $n(s) = 6$

(i) Let E_1 be an event getting an even number

The number of favorable cases to the event of getting an even number is $E_1 = \{2, 4, 6\}$, Hence $n(E_1) = 3$

Probability of getting an even number when die is thrown is $P(E_1) = \frac{3}{6} = \frac{1}{2}$

(ii) Let E_2 be an event getting an odd number, $E_2 = \{1, 3, 5\}$, Hence $n(E_2) = 3$

Probability of getting an even number when die is thrown is $P(E_2) = \frac{3}{6} = \frac{1}{2}$

(iii) Let E_3 be an event getting even or multiple of 3 and $E_3 = \{2, 3, 4, 6\}$, Hence $n(E_3) = 4$

Probability of getting even or multiple of 3 is $P(E_3) = \frac{4}{6} = \frac{2}{3}$

(iv) Let E_4 be an event getting even and multiple of 3 and $E_4 = \{6\}$, Hence $n(E_4) = 1$

Probability of getting even and multiple of 3 is $P(E_4) = \frac{1}{6}$

(v) Let E_5 be an event getting greater than 4 and $E_5 = \{5, 6\}$, Hence $n(E_5) = 2$

Probability of getting greater than 4 is $P(E_5) = \frac{2}{6} = \frac{1}{3}$

(vi) Let E_6 be an event getting composite number and $E_6 = \{4, 6\}$, Hence $n(E_6) = 2$

Probability of getting composite number is $P(E_6) = \frac{2}{6} = \frac{1}{3}$

3. A die is thrown, If the number is odd. What is the probability that it is prime?

Sol: In a die is thrown at a time, If the number is odd, then the possible outcomes are $S = \{1, 3, 5\}$

i.e., The total number of exhaustive events are $n(S) = 3$

Let E be the event of getting a Prime number. i.e., $E = \{3, 5\}$

The number of favorable events are $n(E) = 2$

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{3}$$

4. Two unbiased dice are thrown. Find the probability that (i) Both dice show the same number (ii)

The first dice show '6' (iii) The total of the number on the dice is '8' (iv) The total of the number on

dice is '13' (v) The total of the number on the dice is any number from 2 to 12 (both inclusive).

Sol:

In two dice are thrown at a time then the number of possible outcomes are

$S = \{(1,1), (1,2), \dots, (1,6), (2,1), \dots, (4,1), \dots, (5,1) \dots \dots (6,1) \dots \dots (6,6)\}$ and $n(S) = 36$

(a) Let E_1 be an event getting both dice shows the same number

The number of favorable cases to the event of getting both dice shows the same number is

$E_1 = \{(1,1) (2,2) (3,3) (4,4) (5,5) (6,6)\}$, Hence $n(E_1) = 6$

Probability of getting both dice shows the same number is $P(E_1) = \frac{6}{36} = \frac{1}{6}$

(b) Let E_2 be an event getting the first dice shows the number '6'

The number of favorable cases to the event of getting the first dice shows the number '6' is

$E_2 = \{(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)\}$, Hence $n(E_2) = 6$

Probability of getting the first dice shows the number '6' is $P(E_2) = \frac{6}{36} = \frac{1}{6}$

(c) Let E_3 be an event getting the total of the number on the dice is '8'

The number of favorable cases to the event of getting the total of the number on the dice is '8' is

$E_3 = \{(2,6) (3,5) (4,4) (5,3) (6,2)\}$, Hence $n(E_3) = 5$

Probability of getting the total of the number on the dice is '8' is $P(E_3) = \frac{5}{36}$

(d) Let E_4 be an event getting the sum 13 on the two dice

There is no favorable event of getting the sum 13 when two dice are thrown

$E_4 = \{\}$, Hence $n(E_4) = 0$

Probability of getting the sum 13 on the two dice is $P(E_4) = \frac{0}{36} = 0$

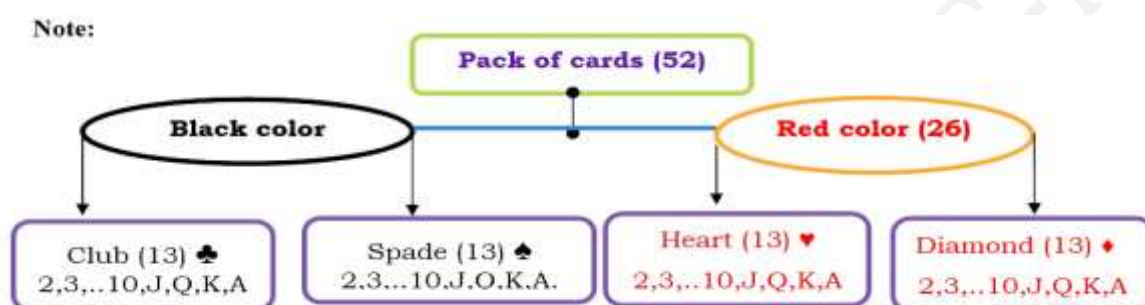
(e) Let E_5 be an event getting the total of the number on the dice is any number from 2 to 12

The number of favorable cases to the total of the number on the dice is any number from 2 to 12 is

$E_5 = \{(1,1) (1,2) \dots (6,6)\}$, Hence $n(E_5) = 36$

Probability of getting the total of the total of the number on the dice is any number from 2 to 12 is

$P(E_5) = \frac{36}{36} = 1$



Note:

- Playing card pack (or) deck contains 52 cards,
- 52 cards are in 4 suits, 13 cards in each suit.
- **Suit(or) Set:** 2,3,4,5,6,7,8,9,10, J, Q, K, A (13 cards)
- **Number Cards:** 2,3,4,5,6,7,8,9,10 (9 cards)
- **Letter Cards:** Jack(J), Queen(Q), King(K), Ace(A) (4 cards)
- **Face (or) Honor Cards:** J, Q, K (3 cards)

5. Four cards are drawn at random from a pack of 52 cards. Find the probability that (i) They are a king, a queen, a jack and an ace. (ii) Two are kings and two are aces. (iii) All are diamonds. (iv) Two are red and two are black. (v) There is one card of each suit. (vi) There are two cards of clubs and two cards of diamonds. (vii) two are letter and two are number cards.

Sol:

Four cards can be drawn from a well shuffled pack of 52 cards in ${}^{52}C_4$ ways. $n(s) = {}^{52}C_4$

(i) 1 king can be drawn out of the 4 kings is ${}^4C_1 = 4$ ways.

Similarly, 1 queen, 1 jack and an ace can each be drawn in ${}^4C_1 = 4$ ways.

The favorable number of cases are ${}^4C_1 * {}^4C_1 * {}^4C_1 * {}^4C_1$.

Hence, required probability = $\frac{{}^4C_1 * {}^4C_1 * {}^4C_1 * {}^4C_1}{{}^{52}C_4}$

- (ii) Required probability = $\frac{4C_2 * 4C_2}{52C_4}$
- (iii) Since 4 cards can be drawn out of 13 cards (since there are 13 cards of diamond in a pack of cards) in $13C_4$ ways, Required probability = $\frac{13C_4}{52C_4}$
- (iv) Since there are 26 red cards (of diamonds and hearts) and 26 black cards (of spades and clubs) in a pack of cards, Required probability = $\frac{26C_2 * 26C_2}{52C_4}$
- (v) Since, in a pack of cards there are 13 cards of each suit, Required probability = $\frac{13C_1 * 13C_1 * 13C_1 * 13C_1}{52C_4}$
- (vi) Required probability = $\frac{13C_2 * 13C_2}{52C_4}$
- (vii) Since there are 16 letter cards and 36 number cards in a pack of 52 cards,
Required probability = $\frac{16C_2 * 36C_2}{52C_4}$

6. Two cards are selected at random from 10 cards numbered 1 to 10. Find the probability that the sum is even if

- (i) The two cards are drawn together.
- (ii) The two cards are drawn one after the other with replacement.
- (iii) The two cards are drawn one after the other without replacement

Sol:

Two cards are drawn at a time out of 10 cards in $10C_2$ ways. $n(s) = 45$

The cards having even numbers = {2, 4, 6, 8, 10}

The cards with odd numbers = {1, 3, 5, 7, 9}

(i) The two cards are drawn together.

Two cards can be selected from 5 even cards in $5C_2$ ways or Two cards can be selected from 5 odd cards in $5C_2$ ways. The number of favorable events are $n(E) = 5C_2 + 5C_2$

Probability that the sum is even if two cards are drawn together $P(E) = \frac{5C_2 + 5C_2}{10C_2} = \frac{20}{45}$

(ii) The two cards are drawn one after the other with replacement.

Suppose we select a card place it back and select one card. This can be happening in $10C_1 * 10C_1$ ways
 $n(s) = 100$

To have an Even Sum, the first card has to be odd and the second card has to be odd (or) the first card has to be even and the second card has to be even.

The number of favorable ways $n(E) = (5C_1 * 5C_1) + (5C_1 * 5C_1) = 50$

Probability that the sum is even if two cards are drawn one after the other with replacement $P(E) = \frac{50}{100} = \frac{1}{2}$

(ii) The two cards are drawn one after the other without replacement.

Suppose we select a card place it not back and select one card. This can be happening in $10P_2$ ways

$$n(s)=90$$

The number of favorable ways $n(E) = (5C_1 * 4C_1) + (5C_1 * 4C_1) = 40$

Probability that the sum is even if two cards are drawn one after the other with replacement $P(E) = \frac{40}{100} = \frac{2}{5}$

7. What is the probability that for a leap year to have 52 Mondays and 53 Sundays?

Sol: A leap year has 366 days. That is 52 weeks and 2 days,

These 2 days can be any one of the following days

(1) Sun&Mon (2) Mon&Tue (3) Tue&Wed (4) Wed&Thu (5) Thu&Fri (6) Fri&Sat (7) Sat&Sun

Number of Exhaustive events $n(S)=7$

Number of favorable events is only Sat&Sun, $n(E)=1$

$$P(E) = \frac{n(E)}{n(S)} = \frac{1}{7}$$

8. What is the chance that a non-leap year should have fifty-three Sundays?

Sol: A non-leap year consists of 365 days, 52 full weeks and one over day.

A non-leap year will consist of 53 Sundays if this over-day is Sunday.

This over-day can be anyone of the possible outcomes: (i) Sunday (ii) Monday (iii) Tuesday (iv)

Wednesday (v) Thursday (vi) Friday (vii) Saturday, ..

7 outcomes in all of these, the number of ways favorable to the required event viz., the over-day being Sunday is 1.

Required probability is $1/7$

9. An urn contains 8 white and 3 red balls. If two balls are drawn at random, find the probability that (i) both are white, (ii) both are red, (iii) one is of each color.

Sol:

Total number of balls in the urn is $8+3=11$.

Since 2 balls can be drawn out of 11 balls in $11C_2$ ways,

Exhaustive number of cases $11C_2=55$

(i) If both the drawn balls are white, they must be selected out of the 8 white balls and this can be done in $8C_2=28$ ways.

Probability that both the balls are white $=28/55$

(ii) If both the drawn balls are red, they must be drawn out of the 3 red balls and this can be done in $3C_2 = 3$ ways.

Hence, the probability that both the drawn balls are red $=3/55$.

(iii) The number of favorable cases for drawing one white ball and one red ball is $8C_1 \cdot 3C_1 = 24$

Probability that one ball is white and other is red $= 24/55$

10. Out of 15 items 4 are not in good condition, 4 are selected at random. Find the probability that (a) all are not good (b) two are not good.

Sol:

Total number of items = 15, Good condition items = 11, Not good condition items = 4

Out of 15 cards 4 can be drawn in $15C_4$ ways. $n(s) = 15C_4$

(a) Number of ways of selecting 4 items which are not good $= 4C_4$

Probability of getting all are not good items $= \frac{4C_4}{15C_4}$

(b) Number of ways of selecting 2 bad items $= 4C_2$

Number of ways of selecting 2 good items $= 11C_2$

Probability of getting two good and two bad items $= \frac{4C_2 \cdot 11C_2}{15C_4} = \frac{99}{455}$

11. A lot consists of 10 good articles, 4 with minor defects and 2 with major defects. Two articles are chosen from the lot at random (without replacement). Find the probability that (i) both are good, (ii) at least 1 is good, (iii) at most 1 is good, (iv) exactly 1 is good, (v) neither has major defects.

Sol: Given that Good articles = 10, Minor defects articles = 2, Major defects articles = 4.

(i) $P(\text{both are good}) = P(GG) = \frac{10C_2}{16C_2} = \frac{3}{8}$

(ii) $P(\text{at least 1 is good}) = P(GB + GG) = \frac{10C_1 \cdot 6C_1 + 10C_2}{16C_2} = \frac{7}{8}$

(iii) $P(\text{at most 1 is good}) = P(GB + BB) = \frac{10C_1 \cdot 6C_1 + 6C_2}{16C_2} = \frac{5}{8}$

(iv) $P(\text{exactly 1 is good}) = P(GB) = \frac{10C_1 \cdot 6C_1}{16C_2} = \frac{1}{2}$

(v) $P(\text{neither has major defects}) = P(\text{non major, non major}) = \frac{14C_2}{16C_2} = \frac{91}{120}$

12. The letters of the word 'article' are arranged at random. Find the probability that the vowels may occupy the even places.

Sol:

The word 'article' contains 7 distinct letters which can be arranged among themselves in $7!$ ways. Hence exhaustive number of cases is $n(s) = 7!$.

In the word 'article' there are 3 vowels (a, e, i) can be arranged in three even places in $3!$ ways.

For each arrangement, the remaining 4 consonants can be arranged in $4!$ ways.

Hence, the number of favourable cases for the vowels to occupy even places is $3! \times 4!$

Probability that the vowels occupy the even places in the word 'article' is $\frac{3! \times 4!}{7!}$

PROBLEMS:

- 13.** In the play of two dice, the thrower loses if his first throw is 2,4, or 12. He wins if his first throw is a 5 or 11. Find the ratio between his probability of losing and probability of winning in the first throw.
- Hint:** Number of favorable cases for getting (a) 2,4 or 12 is $1+3+1=5$; (b) 5 or 11 is $4+2=6$
- 14.** If six dice are rolled, then find the probability that all show different faces.
- 15.** There are 12 cards numbered 1 to 12 cards in a box, if two cards are selected, what is the probability that sum is odd. If (i) With replacement (ii) Without replacement
- 16.** Two digits are selected at random from the digits 1 through 9.
- (i) If the sum is odd, what is the probability that 2 is one of the numbers selected?
- (ii) If 2 is one of the digits selected, what is the probability that the sum is odd?
- 17.** In a box contains 4 granite stones, 5 sand stones, 6 bricks of identical size and shape. Out of them 3 are chosen at random. Find the chance that (i) They belongs to different varieties. (ii) They belongs to same varieties. (iii) They are all granite stones. (iv) 2 of the granite stones. (v) None is Bricks.
- 18.** The letters of the word 'FAILURE' are arranged at random. Find the probability that the consonants may occupy only odd positions.
- 19.** In a random arrangement of the letters of the word 'MATHEMATICS', find the probability that all the vowels come together.
- 20.** Find the probability that in a random arrangement of the letters of the word ASSASSINATION, the four S's, come consecutively
- 21.** Let a coin be tossed. If it shows head we draw a ball from a box containing 3 identical red and 4 identical green balls and if it shows a tail, we throw a die. What is the sample space of experiments?

Answer: $S = \{HR1, HR2, HR3, HG1, HG2, HG3, HG4, T1, T2, T3, T4, T5, T6\}$

(III) Theorems in Probability:

❖ Let A and B be two events in S then

1. At least one of the event A or B occurs = $A \cup B$
2. Both the events A and B occurs = $A \cap B$
3. Neither A nor B occurs = $\bar{A} \cap \bar{B}$
4. Event A occurs and B does not occurs = $A \cap \bar{B}$
5. Exactly one of the events A or B occurs = $(A \cap \bar{B}) \cup (\bar{A} \cap B)$
6. Not more than one of the events A or B occurs = $(A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (\bar{A} \cap \bar{B})$
7. Events are mutually exclusive $A \cap B = 0$
8. Complement of event A is $\bar{A}$

❖ Let A and B & C are three events in S then

1. Only A occurs = $A \cap \bar{B} \cap \bar{C}$
2. Both A and B but not C occurs = $A \cap B \cap \bar{C}$
3. All three events occurs = $A \cap B \cap C$
4. At least one occurs = $A \cup B \cup C$
5. At least two occurs = $(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C) \cup (A \cap B \cap C)$
6. One and no more occurs = $(A \cap \bar{B} \cap \bar{C}) \cup (\bar{A} \cap B \cap \bar{C}) \cup (\bar{A} \cap \bar{B} \cap C)$
7. Two and no more occurs = $(A \cap B \cap \bar{C}) \cup (\bar{A} \cap B \cap C) \cup (A \cap \bar{B} \cap C)$
8. None occurs = $(\bar{A} \cap \bar{B} \cap \bar{C}) = \overline{(A \cup B \cup C)} = 1 - (A \cup B \cup C)$

Some Properties of the Probability Measure:

- a) If ϕ is an empty set (impossible event) then $P(\phi) = 0$
- b) If A is any event in S, then $0 \leq P(A) \leq 1$.
- c) If A be any event of the sample space S, then $P(A') = 1 - P(A)$
- d) For any two events A & B in S Then $P(A \cap B') = P(A) - P(A \cap B)$
- e) For any two events A & B in S Then $P(A' \cap B) = P(B) - P(A \cap B)$
- f) $P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B)$ (De Morgan's Rule)
- g) $P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B)$ (De Morgan's Rule)
- h) For any two events A & B then $P(A \cap B) \leq P(A) \leq P(A \cup B) \leq P(A) + P(B)$
- i) If $A \subset B$, then $P(B - A) = P(B) - P(A)$ and $P(A) \leq P(B)$

State and prove the addition theorem for two events: If A and B are any two events in sample space S, where A & B are not a disjoint event then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

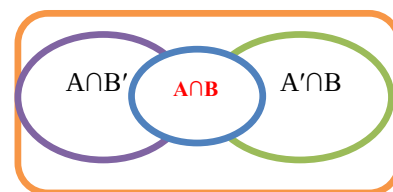
Proof:

We have $A \cup B = A \cup (A' \cap B)$

$$P(A \cup B) = P\{A \cup (A' \cap B)\} = P(A) + P(A' \cap B)$$

$$P(A \cup B) = P(A) + P(A' \cap B) + P(A \cap B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Note: if A, B, C are three events in sample space S and A, B, C are not a disjoint events then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

22. If a card is selected from pack of card, find the probability of getting (a) either spade or queen. (b) either a queen or a king

Sol:

Out of 52 cards one can be drawn in ${}^{52}C_1$ ways. $n(s) = {}^{52}C_1$

Let A be an event that card is spade. $n(A) = 13$

Let B be an event that card is Queen. $n(B) = {}^{4}C_1 = 4$

Let C be an event that card is king. $n(C) = {}^{4}C_1 = 4$

Let $A \cap B$ be an event that card is spade and Queen. $n(A \cap B) = 1$

Let $B \cap C$ be an event that card is Queen and king. $n(B \cap C) = 0$ since B & C are mutually events.

$$P(A) = \frac{n(A)}{n(s)} = \frac{13}{52}$$

$$P(B) = \frac{n(B)}{n(s)} = \frac{4}{52}$$

$$P(C) = \frac{n(C)}{n(s)} = \frac{4}{52}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(s)} = \frac{1}{52}$$

$$P(B \cap C) = \frac{n(B \cap C)}{n(s)} = \frac{0}{52} = 0$$

$$(a) \text{ Probability of getting either spade or queen is } P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52}$$

$$(b) \text{ Probability of getting either queen or king is } P(B \cup C) = P(B) + P(C) - P(B \cap C) = \frac{4}{52} + \frac{4}{52} - 0 = \frac{8}{52}$$

23. A card is drawn from a pack of 52 cards. Find the probability of getting a king or a heart or a red card.

Sol: Out of 52 cards one can be drawn in ${}^{52}C_1$ ways. $n(s) = {}^{52}C_1$

Let A be an event that card is king. $n(A) = {}^{4}C_1$, Let B be an event that card is heart. $n(B) = {}^{13}C_1$

Let C be an event that card is red card. $n(C) = {}^{26}C_1$.

$n(A \cap B) = 1$, $n(B \cap C) = 13$, $n(C \cap A) = 2$, $n(A \cap B \cap C) = 1$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$P(A \cup B \cup C) = \frac{4}{52} + \frac{13}{52} + \frac{26}{52} - \frac{1}{52} - \frac{13}{52} - \frac{2}{52} + \frac{1}{52} = \frac{27}{52}$$

24. From a city 3 newspapers A,B,C are being published. A is read by 20%, B is read by 16%, C is read by 14%, both A and B are read by 8%, both A and C are read by 5%, both B and C are read by 4% and all three read by 2%. What is the percentage of the probability that read at least one paper?

Sol:

$$P(A)=20\%, P(B)=16\%, P(C)=14\%,$$

$$P(A \cap B)=8\%, P(B \cap C)=4\%, P(C \cap A)=5\%, P(A \cap B \cap C)=2\%$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) = 20\% + 16\% + 14\% - 8\% - 5\% - 4\% + 2\%$$

$$P(A \cup B \cup C) = 35\%$$

Percentage of probability that read at least one paper is 35

25. Two dice are thrown. Let A be the event that the sum of the points on the faces is 9. Let B be the event that at least one number is 6. Find (a) $P(A \cap B)$ (b) $P(A \cup B)$ (c) $P(\bar{A} \cup \bar{B})$

Sol:

In two dice are thrown at a time then the number of possible outcomes are $n(s)=36$

Let A be the event of getting sum 9 on the two dice. $n(A)=4$ and $P(A) = \frac{n(A)}{n(s)} = \frac{4}{36}$

Let B be the event of getting at least one number is 6 on the two dice. $n(B)=11$ and $P(B) = \frac{n(B)}{n(s)} = \frac{11}{36}$

Let $A \cap B$ be an event getting sum 9 and at least one number is 6 on the two dice. $n(A \cap B)=2$.

$$P(A \cap B) = \frac{n(A \cap B)}{n(s)} = \frac{2}{36}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{4}{36} + \frac{11}{36} - \frac{2}{36} = \frac{13}{36}$$

$$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - \frac{2}{36} = \frac{34}{36}$$

26. If $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{5}$ then find

a) $P(\bar{A})$

c) $P(A \cup B)$

e) $P(\bar{A} \cap B)$

g) $P(\bar{A} \cap \bar{B})$

b) $P(\bar{B})$

d) $P(A \cap \bar{B})$

f) $P(\bar{A} \cup B)$

h) $P(\bar{A} \cup \bar{B})$

Sol:

a) $P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$

b) $P(\bar{B}) = 1 - P(B) = 1 - \frac{1}{3} = \frac{2}{3}$

c) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{5} = \frac{19}{30}$

d) $P(A \cap \bar{B}) = P(A) - P(A \cap B) = \frac{1}{2} - \frac{1}{5} = \frac{3}{10}$

$$\text{e) } P(\bar{A} \cap B) = P(B) - P(A \cap B) = \frac{1}{3} - \frac{1}{5} = \frac{2}{15}$$

$$\text{f) } P(\bar{A} \cup B) = P(\bar{A}) + P(B) - P(\bar{A} \cap B) = 1 - P(A) + P(B) - P(B) + P(A \cap B) = 1 - \frac{1}{2} + \frac{1}{5} = \frac{6}{10}$$

$$\text{g) } P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - \frac{19}{30} = \frac{11}{30}$$

$$\text{h) } P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B}) = 1 - P(A \cap B) = 1 - \frac{1}{5} = \frac{4}{5}$$

27. If two dice are thrown, what is the probability that their sum is (i) greater than 8 (ii) neither 7 nor 11 (iii) an even number on the first die or a total of 8

Sol:

If two dice are thrown, then $n(s)=36$

$$\text{(i) } P(9 \cup 10 \cup 11 \cup 12) = P(9) + P(10) + P(11) + P(12) = \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{5}{18}$$

$$\text{(ii) Let } A \text{ denote the event of getting the sum of } 7 \text{ then } P(A) = \frac{6}{36}$$

$$B \text{ denote the event of getting the sum of } 11 \text{ then } P(B) = \frac{2}{36}$$

$$P(\text{neither } 7 \text{ nor } 11) = P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$P(\bar{A} \cap \bar{B}) = 1 - \{P(A) + P(B)\} \text{ Where } A \text{ and } B \text{ are mutually exclusive events}$$

$$P(\bar{A} \cap \bar{B}) = 1 - \frac{6}{36} - \frac{2}{36} = \frac{7}{9}$$

(iii) Let } A \text{ be the event of getting an even number on the first die then } A = \{2, 4, 6\} * \{1, 2, 3, 4, 5, 6\} \text{ and } n(A) = 3 * 6 = 18

Let B be the event of getting the sum of 8 then $B = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$ and $n(B) = 5$

$$(A \cap B) = \{(2, 6), (4, 4), (6, 2)\} \text{ and } n(A \cap B) = 3$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{18}{36} + \frac{5}{36} - \frac{3}{36} = \frac{5}{9}$$

Conditional Probability: Let S be a sample space and A and B be any two events in S then

(i) The probability of the occurrence of event A when it is given that B has already occurred is define as $P(A/B) = \frac{P(A \cap B)}{P(B)}$; $P(B) > 0$, This is known as conditional probability of the event A relative to event B.

Similarly, the conditional probability of the event B relative to event A is

$$P(B/A) = \frac{P(A \cap B)}{P(A)}; P(A) > 0$$

Product theorem of probability: Let S be a sample space and A and B be any two events in S, then

$$P(A \cap B) = P(B) * P(A/B) \text{ or } P(A \cap B) = P(A) * P(B/A)$$

Note: Let S be a sample space and A, B and C be three events in S, then

$$P(A \cap B \cap C) = P(A) * P(B/A) * P(C/A \cap B)$$

Properties

1. If A and B are mutually exclusive events $P(B/A) = 0$ since $P(A \cap B) = 0$
2. If $P(A) > P(B)$, $P(A/B) > P(B/A)$
3. If $A \in B$ then $A \cap B = A \Rightarrow P(B/A) = \frac{P(A \cap B)}{P(A)} = 1$

Independent Event: Two or more events are said to be independent events if the happening of the event is not effected by the happening of the other event.

Ex:

1. If we can draw a card from pack of 52 cards and replace it before we draw a second card, then the second draw is independent of first one.
2. If we do not replace the first card before the second draw, then the second draw depend upon first draw.

Note:

- If A and B are independent events then $P(A/B) = P(A)$.

$$\text{Since } P(A/B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A) * P(B)$$

- If A and B are independent events then $P(B/A) = P(B)$

$$\text{Since } P(B/A) = \frac{P(A \cap B)}{P(A)} \Rightarrow P(A \cap B) = P(A) * P(B)$$

- A set of events $\{ A_i \}$ are independent then $P(\cap_{i=1}^n A_i) = \prod_{i=1}^n P(A_i)$
- If A and B are independent events then $P(A \cup B) = 1 - P(\bar{A}) * P(\bar{B})$ i.e; A set of events $\{ A_i \}$ are independent then $P(\cup_{i=1}^n A_i) = 1 - \prod_{i=1}^n P(\bar{A}_i)$
- If the events A and B are independent the events $\bar{A}$ & B, $\bar{A}$ & $\bar{B}$ are also independent

Multiplication theorem for two events: Let S be a sample space and A and B be any two events in S, then

$$P(A \cap B) = \begin{cases} P(A) \cdot P(B/A); & P(A) > 0 \\ P(B) \cdot P(A/B); & P(B) > 0. \\ P(A) \cdot P(B) & \text{if A \& B are independent} \end{cases}$$

28. In a certain college 25% of the students failed in mathematics, 15% failed in chemistry and 10% in both mathematics and chemistry, a student is selected at random

(a) What is the probability that he failed in mathematics if he failed in chemistry.

(b) What is the probability that he failed in chemistry if he failed in mathematics.

Sol:

Let A be an event that student failed in mathematics, $P(A)=25\%$

Let B be an event that student failed in chemistry, $P(B)=15\%$ and $P(A \cap B)=10\%$

(a) Probability that the student failed in mathematics if he failed in chemistry

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{10}{100} \cdot \frac{100}{15} = \frac{2}{3}$$

(b) Probability that the student failed in chemistry if he failed in mathematics

$$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{10}{100} \cdot \frac{100}{25} = \frac{2}{5}$$

29. A die is rolled, if the outcome is an odd number. What is the probability that is prime?

Sol: Sample space $S = \{1, 2, 3, 4, 5, 6\}$

Let A be an event that getting odd number, $A = \{1, 3, 5\}$, $P(A) = 3/6$

Let B be an event that getting prime number, $B = \{2, 3, 5\}$, $P(B) = 3/6$ and $A \cap B = \{3, 5\}$, then $P(A \cap B) = 2/6$

Probability of getting prime number when it is odd number $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{3} \cdot \frac{2}{1} = \frac{2}{3}$

30. A box of fuses contains 20 fuses, of which 5 are defective. If 3 of the fuses are selected at random and removed from the box in succession without replacement, what is the probability that all three fuses are defective?

Sol:

Let A be the event that the first fuse selected is defective. $P(A) = 5/20$

Let B be the event that the second fuse selected is defective. $P(B/A) = 4/19$

Let C be the event that the third fuse selected is defective. $P(C/A \cap B) = 3/18$

The probability that all three fuses selected are defective is

$$P(A \cap B \cap C) = P(A) \cdot P(B/A) \cdot P(C/A \cap B) = \frac{5}{20} \cdot \frac{4}{19} \cdot \frac{3}{18} = \frac{1}{114}$$

31. Two marbles are drawn in succession from a box containing 10 Red, 30 White, 20 Blue and 15 orange marbles, with replacement being made after each draw. Find the probability that (a) both are white (B) first is red and second is white.

Sol:

Total number of marbles $10+30+20+15=75$

Red=10, white=30, blue=20, orange=15

(a) Let A be the event of first drawn marble is white, $P(A) = \frac{{}^{30}C_1}{{}^{75}C_1} = \frac{30}{75}$

Let B be the event of second drawn marble is also white, $P(B) = \frac{{}^{30}C_1}{{}^{75}C_1} = \frac{30}{75}$

Probability that both marbles are white (with replacement) $P(A \cap B) = P(A) \cdot P(B/A)$

$$P(A \cap B) = P(A) \cdot P(B) = \frac{30}{75} * \frac{30}{75} = \frac{4}{25}$$

(b) Let R be an event that the first drawn marble is Red then $P(R) = \frac{{}^{10}C_1}{{}^{75}C_1} = \frac{10}{75}$

Let W be an event that marble is white then $P(W) = \frac{{}^{30}C_1}{{}^{75}C_1} = \frac{30}{75}$

Probability that first marbles is Red and second is white

$$P(R \cap W) = P(R) \cdot P(W / R) = P(R) \cdot P(W) = \frac{10}{75} * \frac{30}{75}$$

32. What is the probability that a family with 2 kids has two boys, given that they have at least one boy? (all possibilities are equally likely).

Sol:

S: all possibilities: {BB, GB, BG, GG}. n(s)=4

X: family has two boys: {BB}, and $P(X)=1/4$

Y: family has at least one boy: {BB, GB, BG}, and $P(Y)=3/4$

$$X \cap Y = \{BB\} \text{ and } P(X \cap Y) = \frac{1}{4}$$

Probability that a family has two boys, given that they have at least one boy $P(X/Y) = \frac{P(X \cap Y)}{P(Y)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$

33. A can hit the target once in 5 shots, B can hit the target 2 in 3 shots, C can hit the target 1 in 4 shots. What is the probability that the two hit the target?

Sol:

Given that $P(A) = \frac{1}{5}$; $P(B) = \frac{2}{3}$; $P(C) = \frac{1}{4}$ and $P(\bar{A}) = \frac{4}{5}$; $P(\bar{B}) = \frac{1}{3}$; $P(\bar{C}) = \frac{3}{4}$

Probability that the two hit the target is

$$\begin{aligned} P\{(A \cap B \cap \bar{C}) \cup (A \cap \bar{B} \cap C) \cup (\bar{A} \cap B \cap C)\} &= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) \\ &= P(A) * P(B) * P(\bar{C}) + P(A) * P(\bar{B}) * P(C) + P(\bar{A}) * P(B) * P(C) \\ &= \left(\frac{1}{5} * \frac{2}{3} * \frac{3}{4}\right) + \left(\frac{1}{5} * \frac{1}{3} * \frac{1}{4}\right) + \left(\frac{4}{5} * \frac{2}{3} * \frac{1}{4}\right) = \frac{1}{4} \end{aligned}$$

34. If $P(A) = \frac{1}{2}$; $P(B) = \frac{1}{3}$; $P(A \cap B) = \frac{1}{5}$ then find (a) $P(A \cup B)$ (b) $P(A/B)$ (c) $P(B/A)$ (d) $P(\bar{A}/B)$ (e) $P(\bar{B}/A)$ (f) $P(\bar{A}/\bar{B})$ (g) $P(\bar{B}/\bar{A})$

Sol:

$$\text{a) } P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{5} = \frac{19}{30}$$

$$\text{b) } P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{5} * \frac{3}{1} = \frac{3}{5}$$

$$\text{c) } P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1}{5} * \frac{2}{1} = \frac{2}{5}$$

$$\text{d) } P(\bar{A}/B) = \frac{P(\bar{A} \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = 3 \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2}{5}$$

$$\text{e) } P(\bar{B}/A) = \frac{P(\bar{B} \cap A)}{P(A)} = \frac{P(A) - P(A \cap B)}{P(A)} = 2 \left(\frac{1}{2} - \frac{1}{5} \right) = \frac{3}{5}$$

$$\text{f) } P(\bar{A}/\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\overline{A \cup B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{1 - P(B)} = \frac{33}{60}$$

$$\text{g) } P(\bar{B}/\bar{A}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{A})} = \frac{P(\overline{A \cup B})}{P(\bar{A})} = \frac{1 - P(A \cup B)}{1 - P(A)} = \frac{11}{15}$$

35. Two dies are thrown; Let A be the event that the sum of the points on the faces is 9. Let B be the event that at least one number is 6, then find (a) $P(A \cup B)$ (b) $P(A' \cap B)$ (c) $P(A \cap B')$ (d) $P(A' \cap B')$ (e) $P(A/B)$ (f) $P(A'/B)$ (g) $P(A'/B')$

36. In a group of 200 students 40 are taking English, 50 are taking mathematics and 12 are taking both.
 (a) If a student is selected at random, what is the probability that the student is taking English?
 (b) A student is selected at random from those taking mathematics. What is the probability that the student is taking English?
 (c) A student is selected at random from those taking English, what is the probability that the student is taking mathematics?

Answer: 0.20, 0.24, 0.3

37. In a box, 100 bulbs are supplied out of which 10 bulbs have defects of type A, 5 bulbs have defects of type B and 2 bulbs have defects of both the type. Find the probability that
 (a) a bulb to be drawn at random has a B type defect under the condition that it has an A type defect,
 (b) a bulb to be drawn at random has no B type defect under the condition that it has no A type defect.

Answer: 0.2, 0.9667

(IV) Bayes theorem and Its applications:

Total Probability: If $E_1, E_2, E_3, \dots, E_n$, are mutually exclusive and exhaustive events of a sample space S , $P(E_i) > 0$ for $i=1$ to n , and for any event A then, $P(A) = \sum P(E_i) \cdot P(A/E_i)$

Bayes' Theorem: If $E_1, E_2, E_3, \dots, E_n$, are mutually exclusive and exhaustive events such that $P(E_i) > 0$ for $i=1, 2, 3, \dots, n$, in a sample space S , and A be any event with $P(A) > 0$ then, $P(E_k / A) = \frac{P(E_k) \cdot P(A/E_k)}{\sum_{i=1}^n P(E_i) \cdot P(A/E_i)}$
for $k=1, 2, 3, \dots, n$

Proof:

Given that $E_1, E_2, E_3, \dots, E_n$, are mutually exclusive and exhaustive events in a sample space S .

$$S = E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n.$$

$$\text{Since } A = A \cap S$$

$$A = A \cap (E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n)$$

$$A = (A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \dots \cup (A \cap E_n)$$

$$P(A) = P\{(A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \dots \cup (A \cap E_n)\}$$

Since $E_1, E_2, E_3, \dots, E_n$ are disjoint events and A be any event, $(A \cap E_1), (A \cap E_2), (A \cap E_3), \dots, (A \cap E_n)$ are disjoint events.

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + P(A \cap E_3) + \dots + P(A \cap E_n)$$

$$P(A) = \sum P(A \cap E_i)$$

$$P(A) = \sum P(E_i) \cdot P(A/E_i)$$

$$\text{Since } P(E_k / A) = \frac{P(A \cap E_k)}{P(A)}$$

$$P(E_k / A) = \frac{P(E_k) \cdot P(A / E_k)}{\sum_{i=1}^n P(E_i) \cdot P(A / E_i)}$$

for $k=1, 2, \dots$

38. There are three boxes. Box-1 contains 7-Red, 3-Black, 5-White, Box-2 contains 9-Red, 2-Black, 5-White, Box-3 contains 10-Red, 5-Black, 5-White balls, one box is chosen and one ball is drawn from it, what is the probability that ball is (a) Red (b) Black (c) White ball.

Sol:

Given that

Box-1 contains 7-Red, 3-Black, 5-White, Total balls=15

Box-2 contains 9-Red, 2-Black, 5-White, Total balls=16

Box-3 contains 10-Red, 5-Black, 5-White, Total balls=20

Let E_1 be an event that box-1 is chosen, $P(E_1)=1/3$

Let E_2 be an event that box-2 is chosen, $P(E_2)=1/3$

Let E_3 be an event that box-3 is chosen, $P(E_3)=1/3$

(a) Let R be an event that red ball is drawn from

Box-1, $P(R/E_1)=7/15$

Box-2, $P(R/E_2)=9/16$

Box-3, $P(R/E_3)=10/20$

The Probability that one box is chosen and one ball is drawn that ball is red

$$P(R) = P(E_1).P(R / E_1) + P(E_2).P(R / E_2) + P(E_3).P(R / E_3)$$

$$P(R) = \frac{1}{3} * \frac{7}{15} + \frac{1}{3} * \frac{9}{16} + \frac{1}{3} * \frac{10}{20} = \frac{47}{90}$$

(b) Let B be an event that Black ball is drawn from

Box-1, $P(B/E_1)=3/15$

Box-2, $P(B/E_2)=2/16$

Box-3, $P(B/E_3)=5/20$

The Probability that one box is chosen and one ball is drawn that ball is Black

$$P(B) = P(E_1).P(B / E_1) + P(E_2).P(B / E_2) + P(E_3).P(B / E_3)$$

$$P(R) = \frac{1}{3} * \frac{3}{15} + \frac{1}{3} * \frac{2}{16} + \frac{1}{3} * \frac{5}{20} = \frac{7}{36}$$

(c) Let W be an event that Black ball is drawn from

Box-1, $P(W/E_1)=5/15$

Box-2, $P(W/E_2)=5/16$

Box-3, $P(W/E_3)=5/20$

The Probability that one box is chosen and one ball is drawn that ball is white

$$P(W) = P(E_1).P(W / E_1) + P(E_2).P(W / E_2) + P(E_3).P(W / E_3)$$

$$P(R) = \frac{1}{3} * \frac{5}{15} + \frac{1}{3} * \frac{5}{16} + \frac{1}{3} * \frac{5}{20} = \frac{51}{180}$$

39. In a bolt factory machines A, B, C manufactures 20%, 30% and 50% of the total their output and 6%,3% and2% are defective. A bolt is drawn at random and found to be defective, find the probability that it is manufacture from (a)Machine-A (b)Machine-B (c)Machine-C

Sol:

Let E_1 be an event that machine A manufactures, $P(E_1)=20\%$

Let E_2 be an event that machine B manufactures, $P(E_2)=30\%$

Let E_3 be an event that machine C manufactures, $P(E_3)=50\%$

Let D be an event those defective bolts of their out put

Machine A, $P(D/E_1)=6\%$

Machine B, $P(D/E_2)=3\%$

Machine C, $P(D/E_3)=2\%$

Total defective bolts $P(D) = P(E_1).P(D/E_1) + P(E_2).P(D/E_2) + P(E_3).P(D/E_3)$

$$P(D) = \frac{20}{100} * \frac{6}{100} + \frac{30}{100} * \frac{3}{100} + \frac{50}{100} * \frac{2}{100} = \frac{31}{1000}$$

(a) Probability that the defective bolt is manufacture from machine A is

$$P(E_1 / D) = \frac{P(E_1) * P(D / E_1).}{\sum_{i=1}^n P(E_i) * P(D / E_i)} = \frac{\frac{20}{100} * \frac{6}{100}}{\frac{31}{1000}} = \frac{12}{31}$$

(b) Probability that the defective bolt is manufacture from machine B is

$$P(E_2 / D) = \frac{P(E_2) * P(D / E_2).}{\sum_{i=1}^n P(E_i) * P(D / E_i)} = \frac{\frac{30}{100} * \frac{3}{100}}{\frac{31}{1000}} = \frac{9}{31}$$

(c) Probability that the defective bolt is manufacture from machine C is

$$P(E_3 / D) = \frac{P(E_3) * P(D / E_3).}{\sum_{i=1}^n P(E_i) * P(D / E_i)} = \frac{\frac{50}{100} * \frac{2}{100}}{\frac{31}{1000}} = \frac{10}{31}$$

40. of the three men the chances that a politician, business man, an educator will be appointed as Vice-Chancellor (V.C) of university are 0.5, 0.3 and 0.2 respectively. Probability that the research is promoted by these people if they are appointed as V.C as are 0.3, 0.7, 0.8 respectively.

(a)Find the probability that research is promoted in the university (b) If research is promoted in the university what is the probability that the V.C is an educator.

Sol:

Let E_1 be an event that the politician is selected as V.C, $P(E_1)=0.5$

Let E_2 be an event that the business man is selected as V.C, $P(E_2)=0.3$

Let E_3 be an event that the educator is selected as V.C, $P(E_3)=0.2$

Let R be an event that the research is promoted as V.C

Politician, $P(R/E_1)=0.3$

Business man, $P(R/E_2)=0.7$

Educator, $P(R/E_3)=0.8$

(a) Probability that the research is promoted in the university

$$P(R) = P(E_1).P(R/E_1) + P(E_2).P(R/E_2) + P(E_3).P(R/E_3)$$

$$P(R) = 0.5 * 0.3 + 0.3 * 0.7 + 0.2 * 0.8 = 0.52$$

(b) When research is promoted in the university then Probability that the V.C is an educator.

$$P(E_3 / R) = \frac{P(E_3) * P(R / E_3)}{\sum_{i=1}^n P(E_i) * P(R / E_i)} = \frac{0.2 * 0.8}{0.52} = \frac{0.16}{0.52} = \frac{4}{13}$$

41. A business man goes to X, Y, Z hotels 20%, 50%, 30% of the times respectively, it is know that 5%, 4% and 8% of the rooms in X, Y, Z hotels have faulty plumbing.

(a) Find the Probability that the business man goes to whole with faulty plumbing.

(b) Find the probability that business man room having faulty plumbing is assigned to hotel Z.

Sol:

Let E_1 be an event that business man goes to hotel X, $P(E_1)=20\%$

Let E_2 be an event that business man goes to hotel Y, $P(E_2)=50\%$

Let E_3 be an event that business man goes to hotel Z, $P(E_3)=30\%$

Let A be an event that rooms having faulty plumbing

Hotel X, $P(A/E_1)=5\%$

Hotel Y, $P(A/E_2)=4\%$

Hotel Z, $P(A/E_3)=8\%$

(a) Total Probability of rooms having faulty plumbing

$$P(A) = P(E_1).P(A/E_1) + P(E_2).P(A/E_2) + P(E_3).P(A/E_3)$$

$$P(A) = \frac{20}{100} * \frac{5}{100} + \frac{50}{100} * \frac{4}{100} + \frac{30}{100} * \frac{8}{100} = \frac{54}{1000}$$

(b) Probability that the business man room having faulty plumbing is assigned to hotel Z

$$P(E_3 / A) = \frac{P(E_3) * P(A / E_3)}{\sum_{i=1}^n P(E_i) * P(A / E_i)} = \frac{\frac{30}{100} * \frac{8}{100}}{\frac{54}{1000}} = \frac{24}{54}$$

42. A toy manufacturing company has categorized all the toys they make into 4 types depending on the battery life of toys (assuming all the toys they manufacture run on battery). It is assumed that the toys from the type 1 category will last for more than 50 hours with a probability of 0.4, while the corresponding probabilities for toys from type 2, 3, and 4 are 0.5, 0.8, and 0.2 respectively. Suppose that 30% of the toys are in category 1, 20% in type 2 category, 10% in

type 3 category, and 40% in type 4 category. Sara went to the toy factory and randomly picked one toy. If the toy did not last for more than 50 hours, what is the probability that she picked the toy from category 4 (assuming the toy will be in use until the battery life is over)?

Sol:

Let E_1 be an event that toy belongs to type-1, $P(E_1) = 30\% = 0.3$

Let E_2 be an event that toy belongs to type-2, $P(E_2) = 20\% = 0.2$

Let E_3 be an event that toy belongs to type-3, $P(E_3) = 10\% = 0.1$

Let E_4 be an event that toy belongs to type-4, $P(E_4) = 40\% = 0.4$

Let L be an event that battery last for more than 50 hours

Type-1, $P(L/E_1) = 0.4$

Type-2, $P(L/E_2) = 0.5$

Type-3, $P(L/E_3) = 0.8$

Type-4, $P(L/E_4) = 0.2$

$$P(L^c) = \sum_{i=1}^n P(E_i) * P(L^c / E_i)$$

$$P(L^c) = P(E_1).P(L^c / E_1) + P(E_2).P(L^c / E_2) + P(E_3).P(L^c / E_3) + P(E_4).P(L^c / E_4)$$

$$P(L^c) = (0.3 * 0.6) + (0.2 * 0.5) + (0.1 * 0.2) + (0.4 * 0.8) = 0.62$$

$$P(E_4 / L^c) = \frac{P(E_4) * P(L^c / E_4)}{\sum_{i=1}^n P(E_i) * P(L^c / E_i)} = \frac{0.4 * 0.8}{0.62} = 0.5161$$

43. A box contains 3 blue and 2 red marbles while another box contains 2 blue and 5 red marbles. A marble drawn at random from one of the boxes. Find the probability that it is a blue marble.

Answer: 31/70

44. An urn contains 10 white and 3 black balls, while another urn contains 3 white and 5 black balls. Two balls are drawn from the first urn and put into the second urn and then a ball is drawn from the later. What is the probability that it is a white ball?

Answer: 59/130

45. In a certain assembly plant, three machines, B1, B2 and B3, make 30%, 45% and 25%, respectively, of the products. It is known from the past experience that 2%, 3% and 2% of the products made by each machine respectively are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?

Answer: 0.0245

46. Suppose that the population of a certain city is 40% male and 60% female. Suppose also that 50% of the males and 30% of the female smoke. Find the probability that a smoker is male.

Answer: 10/19

47. A microchip company has two machines that produce the chips. Machine-I produces 65% of the chips, but 5% of its chips are defective. Machine-II produces 35% of the chips, but 15% of its chips are defective. A chip is selected at random and found to be defective. What is the probability that it came from Machine-I?

Answer: 0.3824

48. In a bolt factory, three machines A, B and C manufacture 25%, 35% and 40% of the total product respectively. out of these outputs 5%, 4% and 2% respectively, are defective bolts. A bolt is picked up at random and found to be defective. What are the Probabilities that it was manufactured by machine A, B and C?

Answer: 0.3623, 0.4057, 0.2318

49. A company has two plants to manufacture hydraulic machine. Plant I manufacture 70% of the hydraulic machines and plant II manufactures 30%. At plant I, 80% of hydraulic machines are rated standard quality and at plant II, 90% of hydraulic machine are rated standard quality. A machine is picked up at random and is found to be of standard quality. What is the chance that it has come from plant I?

Answer: 0.6747

50. There are two boxes A and B containing 4 white, 3 red and 3 white, 7 red balls respectively. A box is chosen at random and a ball is drawn from it, if the ball is white, find the probability that it is from box A.

Answer: 40/61

51. If proposed medical screening on a population, the probability that the test correctly identifies someone with illness as positive is 0.99 and the probability that test correctly identifies someone

without illness as negative is 0.95. The incident of illness in general population is 0.0001. You take the test; the result is positive then what is the probability that you have illness?

Answer: 0.002

52. In producing screws, let A mean “screw too slim” and B “screw too short”. Let $p(A) = 0.1$ and let the conditional probability that a slim screw is also too small be $P(B/A) = 0.2$. What is the probability of screw, we pick randomly from the lot produced, will be both too slim and too short?

Answer: 0.02

53. Three boxes contain 10%, 20%, and 30% of defective finger joints. A finger joint is selected at random which is defective. Determine probability that it comes from i) 1st box ii) 2nd box iii) 3rd box

Answer: 1/6, 1/3, 1/2

54. Suppose there are three chests each having two drawers. The first chest has a gold coin in each drawer, the second chest has a gold coin in one drawer and a silver coin in the other drawer and the third chest has a silver coin in each drawer. A chest is chosen at random and a drawer opened. If the drawer contains a gold coin, what is the probability that the other drawer also contains a gold coin?

Answer: 2/3

55. In a railway reservation office, two clerks are engaged in checking reservation forms. On an average, the first clerk checks 55% of the forms, while the second does the remaining. The first clerk has an error rate of 0.03 and second has an error rate of 0.02, A reservation form is selected at random from the total number of forms checked during a day, and is found to have an error. Find the probability that it was checked (i) by the first (ii) by the second clerk.

56. In a class of 75 students, 15 were considered to be very intelligent, 45 as medium and the rest of them are below average. The probability that a very intelligent student fails in a viva – voice examination is 0.005, the medium student failing has a probability 0.05; and the corresponding probability for a below average student is 0.15. If a student is known to have passed the viva – voice examination, what is the probability that he is below average?

57. A company has two plants to manufacture scooters. Plant I manufactures 80 percent of the scooters and plant II manufactures 20 percent. At plant I, 85 out of 100 scooters are rated standard quality or better. At plant II, 65 only out of 100 scooters are rated standard quality or better. What is the probability that scooter selected at random came from plant I if it is known that the scooter is of standard quality?

What is the probability that the scooter came from plant II, if it is known that the scooter is of standard quality?